

# Question and method

Does averaging over valid contact serializations improve contact prediction?

Use the current decontaminated exp232 recipe for two fresh, matched arms. Soften contact endpoints with exact generator probabilities. Keep sequence statements, inputs, positions, attention and packing unchanged.

# Implementation checks

60 CPU tests pass; the full-size accelerator loss/gradient test passes. An independent oracle checks 8192 positions, 2048 hidden, 2845 vocabulary. Two virtual CPU devices pass accumulation, resume and SkipStep migration tests. Both arms train for three GPU steps, save native state, validate and export. Ordinary Marifold inference checks the exported model and tokenizer.

# What these checks do not show

There is no accuracy result yet; the corpora were not independently re-audited. A full 1.47B, 32-H100 pilot completed 32 updates at about 3.5 seconds/update. It saved native state and HF weights/tokenizer, but exposed missing validation. The cache configuration is corrected and tested with actual validation reads. The corrected pilot passed: 32 updates, finite validation CE 6.3678, saved state. The production driver continues that exact checkpoint on 32 H100s. The soft arm is launched; matched CE and accuracy evaluation remain future work. It reached step 55520 before one worker exited 139; a frozen-source relaunch from the complete step-55265 recovery checkpoint is queued for the same 32-H100 setup.

# Readout

Compare natural-protein eval-val R-precision at matched exposure and compute. Use identical rollout budgets and paired protein-level uncertainty. Keep ordinary validation CE comparable; report soft CE with entropy and KL. Confirm the fresh CE baseline and repeat promising results with another seed.

# Learning-rate continuation result

Fork permanent soft checkpoint step 14520 at LR 0.001, 0.0015 and 0.002. Keep Adam moments, RNG, data position, batch 128 and 32-H100 geometry fixed. Run 5,000 further updates per branch; leave the original production run running. Track ordinary and contact-endpoint CE, gradient norms and clipping decisions. Native optimizer fork tests and a stock GPU diagnostic smoke pass. The parent full-state checkpoint manifest passes restore preflight. The distributed pilot passed both CE metrics, native state and HF/tokenizer saves. All three completed 5,000 updates on 32 Reno H100s each; all gangs exited. First-update loss/gradients match; update norms scale by 1x, 1.5x and 2x. Final document CE: 3.350010 / 3.360617 / 3.360499 for LR 0.001 / 0.0015 / 0.002. Final endpoint CE: 4.351321 / 4.367665 / 4.358667; lower is better for both. Keep 0.001: it leads both metrics at every shared evaluation point. This is one seed and an abrupt mid-training LR change; no contact accuracy yet. Final native manifests and HF/tokenizer file presence verified, not full reloads. Soft-vs-one-hot success still needs an appropriately tuned CE control.

# lr\_validation.png

Ordinary one-hot validation CE, lower is better. Each panel uses a fixed evaluation set across the three branches. All start from the same full-state checkpoint; the control also includes its two-update pilot.

